

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <title>Lab 7.3 D3 Stacked Bar Chart</title>
6   <script src="https://d3js.org/d3.v7.min.js"></script>
7   <style>
8     svg {
9       font-family: sans-serif;
10    }
11    .bar text {
12      fill: white;
13      font-size: 10px;
14    }
15  </style>
16 </head>
17 <body>
18   <h2>Lab 7.3 - D3 Stacked Bar Chart</h2>
19   <svg width="300" height="300"></svg>
20
21   <script>
22     const dataset = [
23       { apples: 5, oranges: 10, grapes: 22 },
24       { apples: 4, oranges: 12, grapes: 28 },
25       { apples: 2, oranges: 19, grapes: 32 },
26       { apples: 7, oranges: 23, grapes: 35 },
27       { apples: 23, oranges: 17, grapes: 43 }
28     ];
29
30     const keys = ["apples", "oranges", "grapes"];
31
32     const w = 300;
33     const h = 300;
34     const padding = 30;
35
36     const svg = d3.select("svg")
37       .attr("width", w)
38       .attr("height", h);
39
40     // === Create stack series ===
41     const stack = d3.stack()
42       .keys(keys);
43
44     const series = stack(dataset); // array of 3 layers (apples, oranges, grapes)
45
46     // === Scales ===
47     const xScale = d3.scaleBand()
48       .domain(d3.range(dataset.length))
49       .range([padding, w - padding])
50       .padding(0.1);
51
52     const yScale = d3.scaleLinear()
53       .domain([0, d3.max(dataset, d =>
54         d.apples + d.oranges + d.grapes)])
55       .range([h - padding, padding]);
56
57     // === Color scale ===
58     const color = d3.scaleOrdinal()
59       .domain(keys)
60       .range(d3.schemeCategory10);
61
62     // === Create groups for each layer ===
63     svg.selectAll("g.layer")
64       .data(series)
65       .enter()
66       .append("g")
67       .attr("class", "layer")
68       .attr("fill", (d, i) => color(keys[i]))
69       .selectAll("rect")
70       .data(d => d)
71       .enter()
72       .append("rect")
73       .attr("x", (d, i) => xScale(i))
74       .attr("y", d => yScale(d[1]))
75       .attr("height", d => yScale(d[0]) - yScale(d[1]))
76       .attr("width", xScale.bandwidth());
77
78     // === Axes ===
79     const yAxis = d3.axisLeft(yScale).ticks(5);
80
81     svg.append("g")
82       .attr("transform", `translate(${padding}, 0)`)
83       .call(yAxis);
84   </script>
85 </body>
86 </html>

```